Abstract Algebra | Quiz 7 | October 21, 2026

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1. Write an addition table for the group $\mathbb{Z}_6/\langle 3\rangle$.

$0+\langle 3\rangle = \{0, 3\}$
$1+\langle 3\rangle = \{1, 4\}$
$2+\langle 3\rangle = \{2\ 5\}$

} cosets (elements of $\mathbb{Z}_6/\langle 3\rangle$)

+	$0+\langle 3\rangle$	$1+\langle 3\rangle$	$2+\langle 3\rangle$
$0+\langle 3\rangle$	$0+\langle 3\rangle$	$1+\langle 3\rangle$	$2+\langle 3\rangle$
$1+\langle 3\rangle$	$1+\langle 3\rangle$	$2+\langle 3\rangle$	$0+\langle 3\rangle$
$2+\langle 3\rangle$	$2+\langle 3\rangle$	$0+\langle 3\rangle$	$1+\langle 3\rangle$

2. Recall that the **center** of a group G is the subgroup $Z(G) = \{x \in G : xg = gx,\ \forall g \in G\} \subseteq G$. Prove that this subgroup is a *normal* subgroup of G.

Proof We must show $gZ(G) = Z(G)g$ for any $g \in G$. So let $g \in G$.

Observe that

$$gZ(G) = \{gx : x \in Z(G)\}$$
$$= \{xg : x \in Z(G)\} \quad (gx = xg \text{ because } x \in Z(G))$$
$$= Z(G)g.$$

Because $gZ(G) = Z(G)g$, it follows that $Z(G)$ is a normal subgroup, by definition of a normal subgroup. ∎